

Chandan Kumar Singh

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Summary

Software Engineer with 1+ year of experience building scalable, low-latency microservices using Java, Spring Boot, Kafka, and AWS; delivered 40% latency reduction and 60% overhead cut through high-ownership engineering. Proven track record designing production-grade APIs, securing distributed systems, and automating workflows. Strong DSA background (1000+ problems solved, CodeChef Global Rank 14).

Experience

Tata Consultancy Services

Software Engineer — Java, Spring Boot, Microservices

Jan 2025 – Present

- Designed and owned RESTful APIs for business-critical banking workflows, cutting average response latency by 40% and boosting platform scalability by 50% through service-level optimizations and efficient resource utilization.
- Architected a centralized API Gateway with secret-key authentication and JWT token-reuse strategy, reducing inter-service overhead by 60% and enhancing distributed system reliability and security.
- Engineered secure integrations with external banking systems using AES-256 encryption and defensive patterns against MITM, replay, and request-tampering attacks, achieving zero security incidents in production.
- Owned end-to-end delivery of internal automation solutions, eliminating ~64% of manual operational effort and accelerating team release velocity via CI/CD pipelines.
- Engineered a high-throughput bulk-upload pipeline for concurrent ingestion of thousands of records per request, increasing system throughput by 35% while maintaining strict data consistency.

Projects

Clothify Shopping Application | Java, Spring Boot, Hibernate, PostgreSQL, React.js, gRPC, Kafka, Docker, Kubernetes, AWS

- Built a scalable e-commerce backend using Java, Spring Boot, Hibernate, PostgreSQL, React.js, gRPC, Kafka, Docker, Kubernetes, and AWS; implemented API Gateway routing, Eureka service discovery, and domain-driven microservices architecture.
- Engineered low-latency gRPC + Kafka-based asynchronous communication with JWT security; containerized services and deployed on Kubernetes (AWS EC2) via Jenkins CI/CD, supporting 100+ concurrent users and reducing latency by 30%.

QuantPulse AI | Python, Java, Spring Boot, FastAPI, React.js, XGBoost, Random Forest, FinBERT, PostgreSQL, Redis, WebSockets

- Developed an ensemble ML engine combining XGBoost, Random Forest, and Logistic Regression on 60+ technical indicators with FinBERT-based news sentiment, improving prediction accuracy over single-model baselines.
- Architected a real-time streaming stack (Spring Boot + FastAPI + React + WebSocket) with Redis caching and resilient Groww API integration using exponential backoff, enabling tick-by-tick market data ingestion.

Technical Skills

Languages: Java, Python, C++, SQL, JavaScript, TypeScript

Frameworks: Spring Boot, Spring Security, Hibernate, Microservices, gRPC, Kafka, REST API, FastAPI, React.js

Databases: MySQL, PostgreSQL, Redis, FAISS, ChromaDB

Cloud: AWS (EC2, S3), Docker, Kubernetes, Jenkins

Tools: Git, GitHub, SVN, Jenkins, Docker, Kubernetes, NumPy, Pandas, Scikit-learn, HuggingFace, Transformers, LangChain

Methodologies: Agile, CI/CD, DevOps

Education

Motihari College of Engineering, Motihari

B.Tech in Computer Science and Engineering (CSE)

Oct 2020 – Aug 2024